

Euler's number

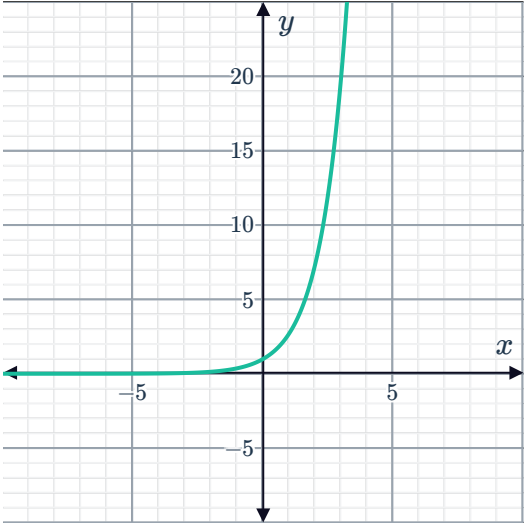


1

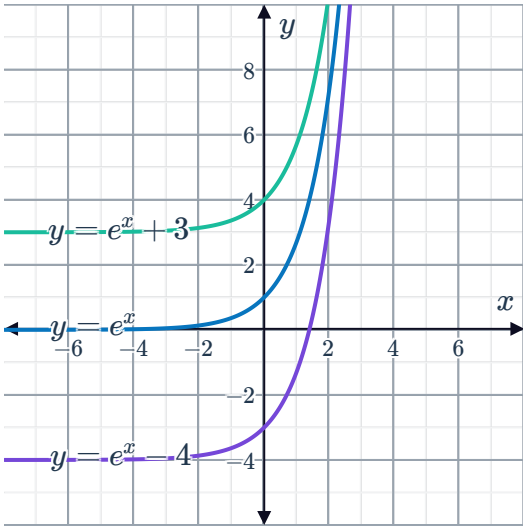
a

x	-3	-2	-1	0	1	2	3
$f(x)$	0.05	0.14	0.37	1.00	2.72	7.39	20.09

b



2



3

- a 54.5982
- b 0.3679
- c 1.2214
- d 8.2436
- e 1.4715
- f 0.0163

The natural logarithm

4

- a 4.5422
- b 2.1701
- c 17.4968
- d 6.4457



5 $x = 3.43$

6 5

7 Less than 1

8

a 3.5

b 4

c 6

d -2

9

a $0 = \ln 1$

b $x = \ln 3$

c $\ln \sqrt{e} = \frac{1}{2}$

d $p = \ln q$

Applications

10

a 3000

b $k = \frac{1}{2} \ln \left(\frac{8}{3} \right)$

c $t = 2$

11 $t = 1.39$

12

a 643

b $t = 8$